1. Several different antigens can be detected in blood tests. The following four traits were tested for each individual shown:

   ABO type (I^A and I^B codominant, i recessive)
   Rh type (Rh^+ dominant to Rh^-)
   MN type (M and N codominant)
   $Xg^{(a)}$ type ($Xg^{(a+)}$ dominant to $Xg^{(a-)}$)

   All of these blood type genes are autosomal, except for $Xg^{(a)}$, which is X linked.

Mother	AB	Rh^-	MN	$Xg^{(a+)}$
Daughter	A	Rh^+	MN	$Xg^{(a-)}$
Alleged father 1	AB	Rh^+	M	$Xg^{(a+)}$
Alleged father 2	A	Rh^-	N	$Xg^{(a-)}$
Alleged father 3	B	Rh^+	N	$Xg^{(a-)}$
Alleged father 4	O	Rh^-	MN	$Xg^{(a-)}$

   a. Which, if any, of the alleged fathers could be the real father? Why?

   b. Would your answer to part *a* change if the daughter had Turner syndrome (the abnormal phenotype seen in XO individuals)? If so, how?

a. If parents are Rh^-, then children can't be Rh^+, so we can exclude father 2, 4

Suppose 1 could be the real father:

ABO type: I^AI^B × I^Ai → I^AI^A, I^Ai ✓

Rh type: Dd × dd → Dd, dd (DD or Dd) ✓

MN type: L^ML^N × L^ML^N → L^ML^M, L^ML^N, L^NL^N ✓

$Xg^{(a)}$ type: record $Xg^{(a+)}$ as A, $Xg^{(a-)}$ as a.

X^AY × X^AX^a → X^AX^A, X^AY, X^aY, X^aX^a (X^AX^A or X^AX^a)

X^aY can only be X^AY ✗

So he's not the real father.

Suppose 3 could be the real father:

ABO type: I^Bi × I^Ai → I^AI^B, I^Bi, I^Ai, ii ✓

Rh type: Dd × dd → Dd, dd (DD or Dd) ✓

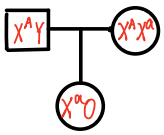
MN type: L^NL^N × L^ML^N → L^ML^N, L^NL^N ✓

$Xg^{(a)}$ type: X^aY × X^AX^a → X^AY, X^aY, X^AX^a, X^aX^a ✓

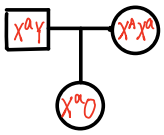
According to what I discussed above, the real father is father 3.

b. if daughter has XO syndrome, then $Xg^{(a)}$ genotype should be rediscussed.

father 1:

$Xg^{(a)}$ type:  ✓

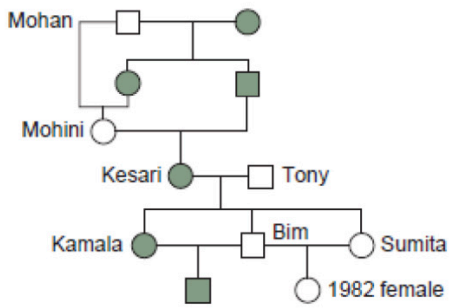
father 3:

$Xg^{(a)}$ type:  ✓

So father 1, 3 can both be the possible real father.

2. The ancestry of a white female tiger bred in a city zoo is depicted in the pedigree following part (e) of this problem. White tigers are indicated with unshaded symbols. (As you can see, there was considerable inbreeding in this lineage. For example, the white tiger Mohan was mated with his daughter.) In answering the following questions, assume that "white" is determined by allelic differences at a single gene and that the trait is fully penetrant. Explain your answers by citing the relevant information in the pedigree.

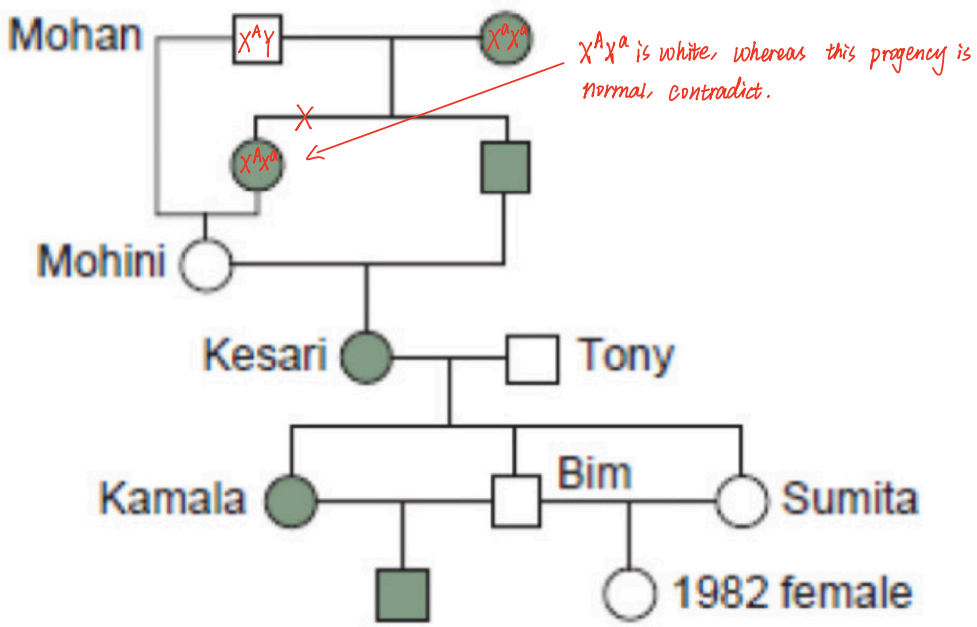
a. Could white coat color be caused by a Y-linked allele?
b. Could white coat color be caused by a dominant X-linked allele?
c. Could white coat color be caused by a dominant autosomal allele?
d. Could white coat color be caused by a recessive X-linked allele?
e. Could white coat color be caused by a recessive autosomal allele?



Indicate "white": dominant "A", recessive "a".

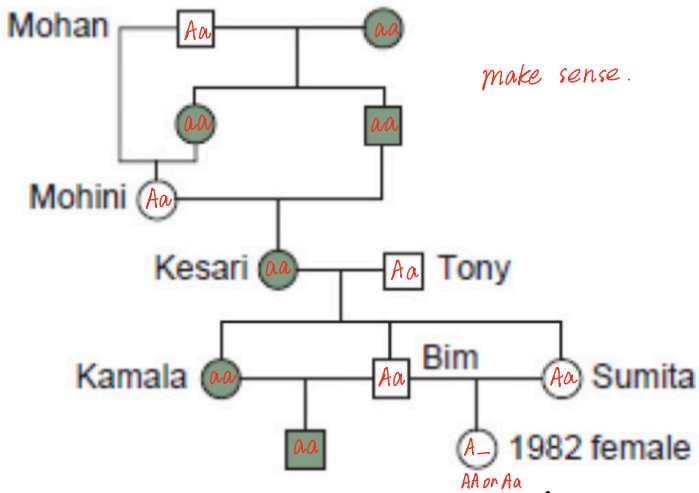
a. If it's Y-linked allele, then all the male should be white, contradicted, so it can't be Y-linked allele. X

b. Suppose it's a dominant X-linked allele.



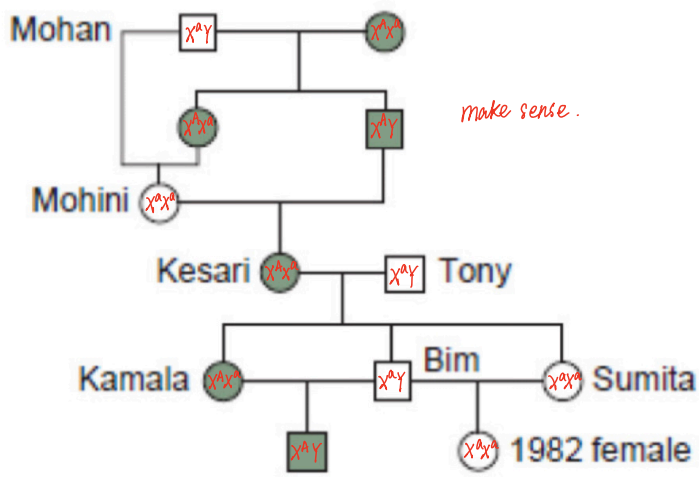
X

c. Suppose it's a dominant autosomal allele. ✓



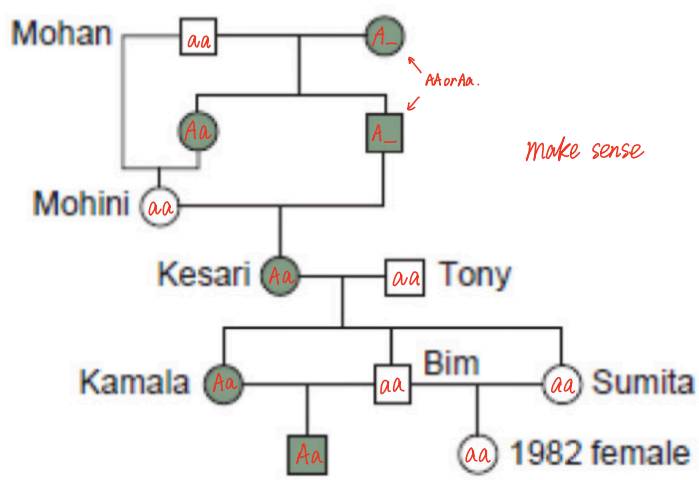
make sense.

d. Suppose it's a recessive X-linked allele. ✓



make sense.

e. Suppose it's a recessive autosomal allele. ✓



make sense

3. The Xg cell-surface antigen is coded for by a gene located on the X chromosome. No equivalent gene exists on the Y chromosome. Two codominant alleles of this gene have been identified: *Xg1* and *Xg2*. A woman of genotype *Xg2/Xg2* bears children with a man of genotype *Xg1/Y*, and they produce a son with Klinefelter syndrome of genotype *Xg1/Xg2/Y*. Using proper genetic terminology, briefly explain how this individual was generated. In which parent and in which meiotic division did the mistake occur?

overall:

$Xg_1/Y \times Xg_2/Xg_2$

↓

$Xg_1/Xg_2/Y$

explain:

父亲在产生配子时，减数第一次分裂未将 Xg_1 与 Y 分离。

